

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

FetchKart – Multi-Vendor E-Commerce Platform

Prepared By

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1. Introduction

FetchKart is a full-stack multi-vendor e-commerce platform developed using HTML, CSS, JavaScript, PHP, and MySQL. The platform is designed to simulate a real-world online marketplace where customers can purchase products, sellers can manage inventory, and administrators can monitor platform activity.

The system integrates secure authentication, payment processing, order management, and responsive user interface design to deliver a seamless shopping experience across multiple user roles.

2. Scope of the System

The primary objective of FetchKart is to provide a scalable and user-friendly e-commerce solution with multiple integrated modules.

The platform supports:

- Product browsing and purchasing
- Seller inventory and product management
- Administrative monitoring and analytics
- Coupon and discount management
- Secure checkout and payment workflows
- Order tracking and delivery monitoring

The system emphasizes responsive UI design, backend security, and modular architecture.

3. Technologies Used

Technology	Purpose
HTML5	Frontend Structure

Technology	Purpose
CSS3	Styling and Responsive Design
JavaScript	Client-Side Functionality
PHP	Backend Logic and APIs
MySQL	Database Management
Razorpay API	Payment Gateway Integration
Cloudflare Turnstile	CAPTCHA Verification

4. User Roles

Customer

Customers can:

- Browse and search products
- Add products to the cart
- Apply discount coupons
- Place orders
- Track delivery progress

Seller

Sellers can:

- Add and manage products
- Monitor inventory levels
- Update stock information
- Access seller dashboard analytics

Administrator

Administrators can:

- Manage users and products
 - Monitor orders and platform activity
 - Create and manage coupons
 - Access platform analytics dashboards
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5. Functional Requirements

Authentication System

The platform includes a secure authentication system with:

- User registration and login
- Session management
- Password hashing
- CAPTCHA verification
- Role-based access control

Product Management System

The product management module allows sellers to:

- Add, edit, and delete products
- Upload product images
- Configure pricing and categories
- Manage stock quantities

Shopping Cart System

The shopping cart module supports:

- Adding and removing products
- Updating item quantities
- Dynamic cart total calculation
- Persistent cart management

Checkout and Payment System

The checkout workflow includes:

- Shipping information forms
- Payment method selection
- Coupon application
- Order confirmation
- Razorpay payment integration

Supported Payment Methods

- UPI
- Razorpay
- Cash on Delivery (COD)

Coupon Management System

The coupon management module supports:

- Percentage-based discounts
- Fixed discount values
- Expiry date validation
- Minimum order amount verification

Order Tracking System

The tracking system provides:

- Delivery progress updates
 - Estimated delivery information
 - Order history management
 - Shipment status visualization
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6. Non-Functional Requirements

Security Requirements

The platform implements multiple security mechanisms, including:

- Password hashing
- Session authentication
- Input validation
- CAPTCHA protection
- Role-based authorization

Performance Requirements

The system is optimized for:

- Fast page rendering
- Efficient database queries
- Smooth user interaction
- Responsive layouts across devices

Usability Requirements

The user interface is designed to provide:

- Easy navigation
- Responsive design
- Dark and light mode support
- Accessible shopping workflows

7. System Architecture

Frontend Components

The frontend interface consists of:

- Landing Page
- Shop Page
- Cart Page
- Checkout Page
- Orders Page
- Seller Dashboard
- Admin Dashboard

Backend Components

The backend system handles:

- User authentication
- Product APIs
- Coupon validation
- Order processing
- Database communication

Database Structure

Table Name	Purpose
users	Stores user information
products	Stores product details
orders	Stores order records
cart	Stores cart items
coupons	Stores coupon data

8. Testing and Validation

The platform was tested for:

- Authentication functionality
- CRUD operations
- Payment workflows
- Database interactions
- Responsive UI behavior

- Form validation and error handling
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9. Future Enhancements

Potential future upgrades include:

- Product review and rating systems
 - AI-based product recommendations
 - Mobile application support
 - Multi-language support
 - Advanced analytics dashboards
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10. Conclusion

FetchKart demonstrates the implementation of a modern full-stack e-commerce platform using frontend and backend web technologies.

The project successfully integrates responsive user interface design, secure authentication mechanisms, payment gateway integration, database management, and role-based functionality into a single scalable system.

The platform reflects practical knowledge of full-stack web development concepts, including CRUD operations, API integration, database handling, authentication systems, and responsive UI/UX design.

End of Report